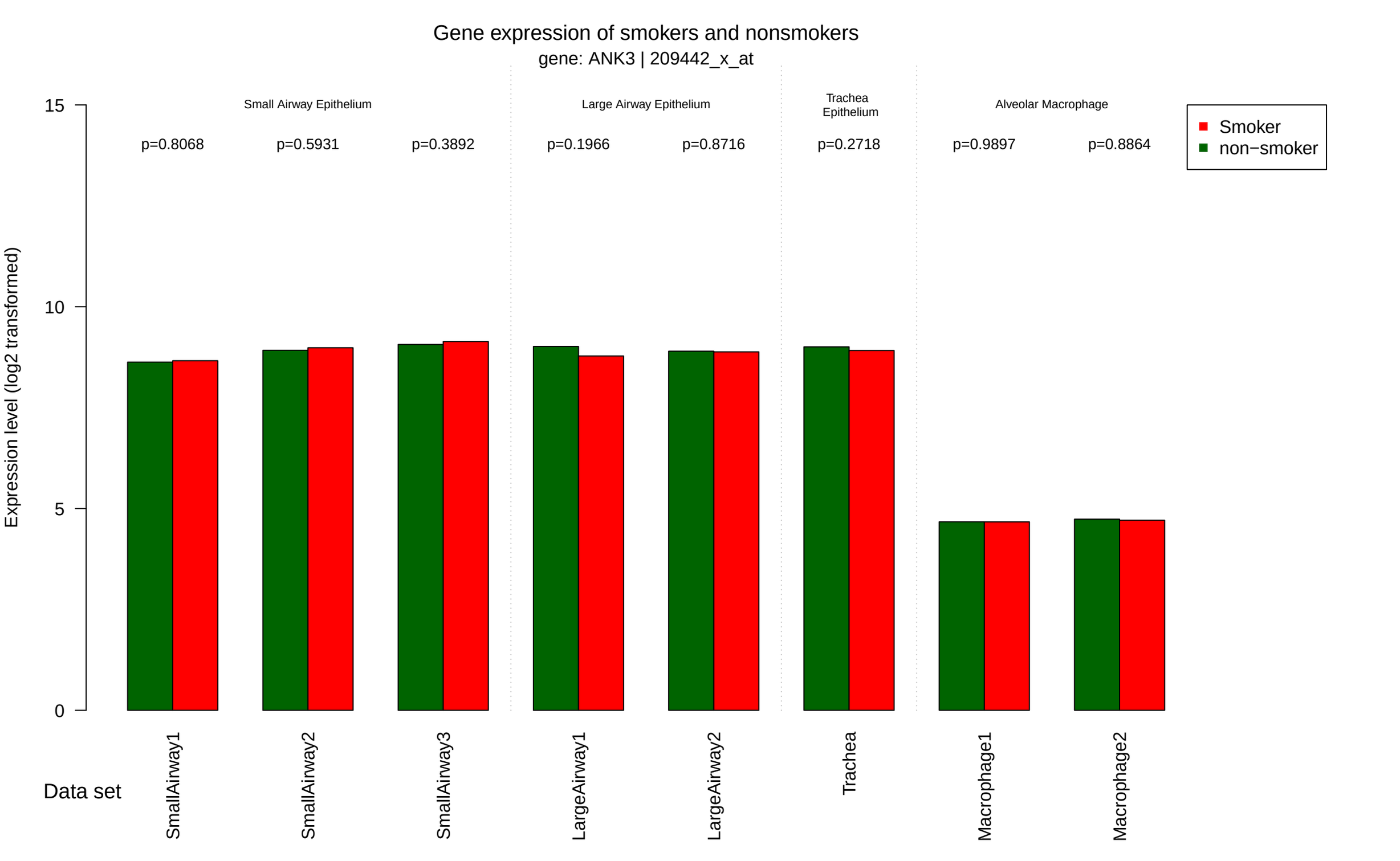
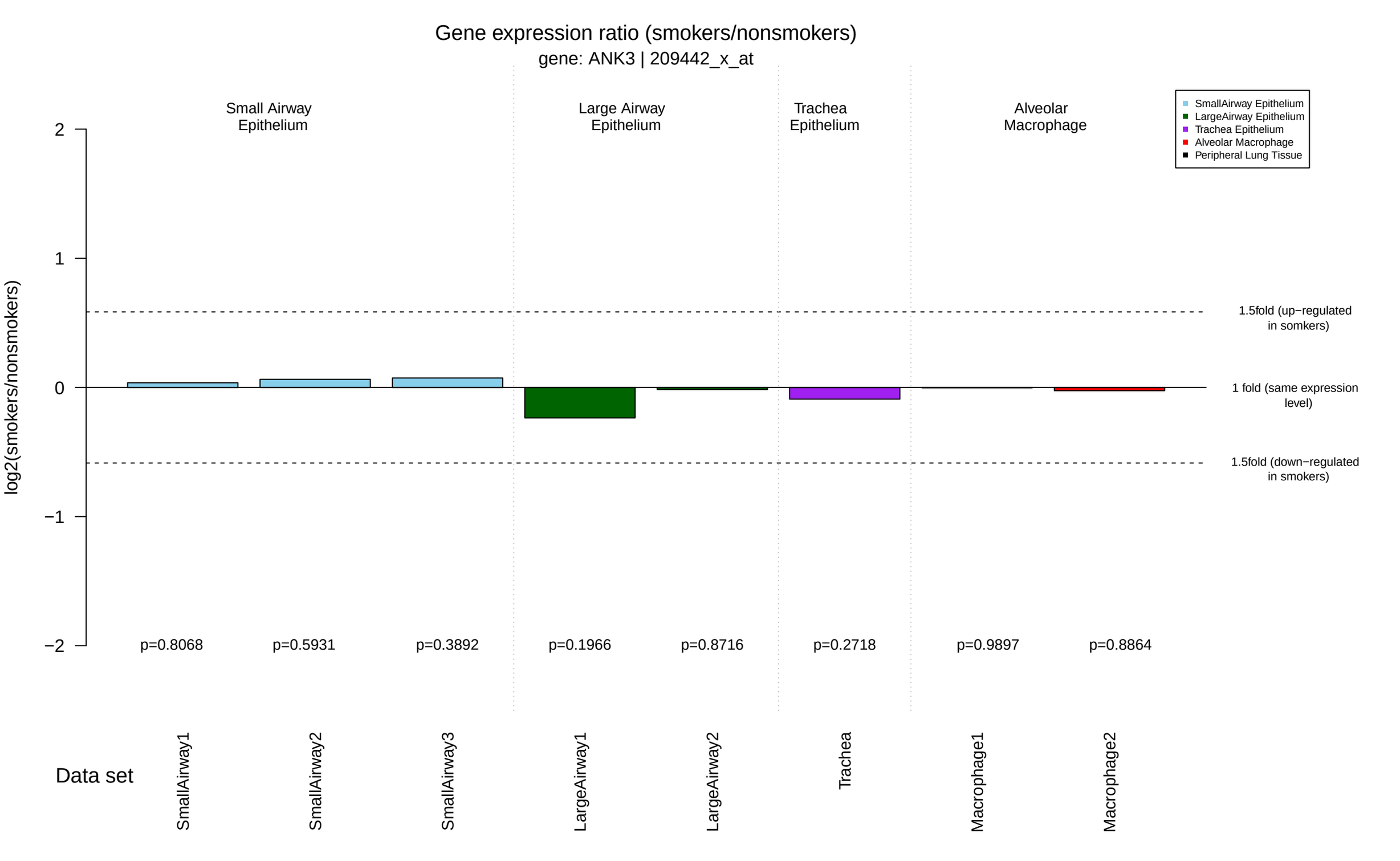
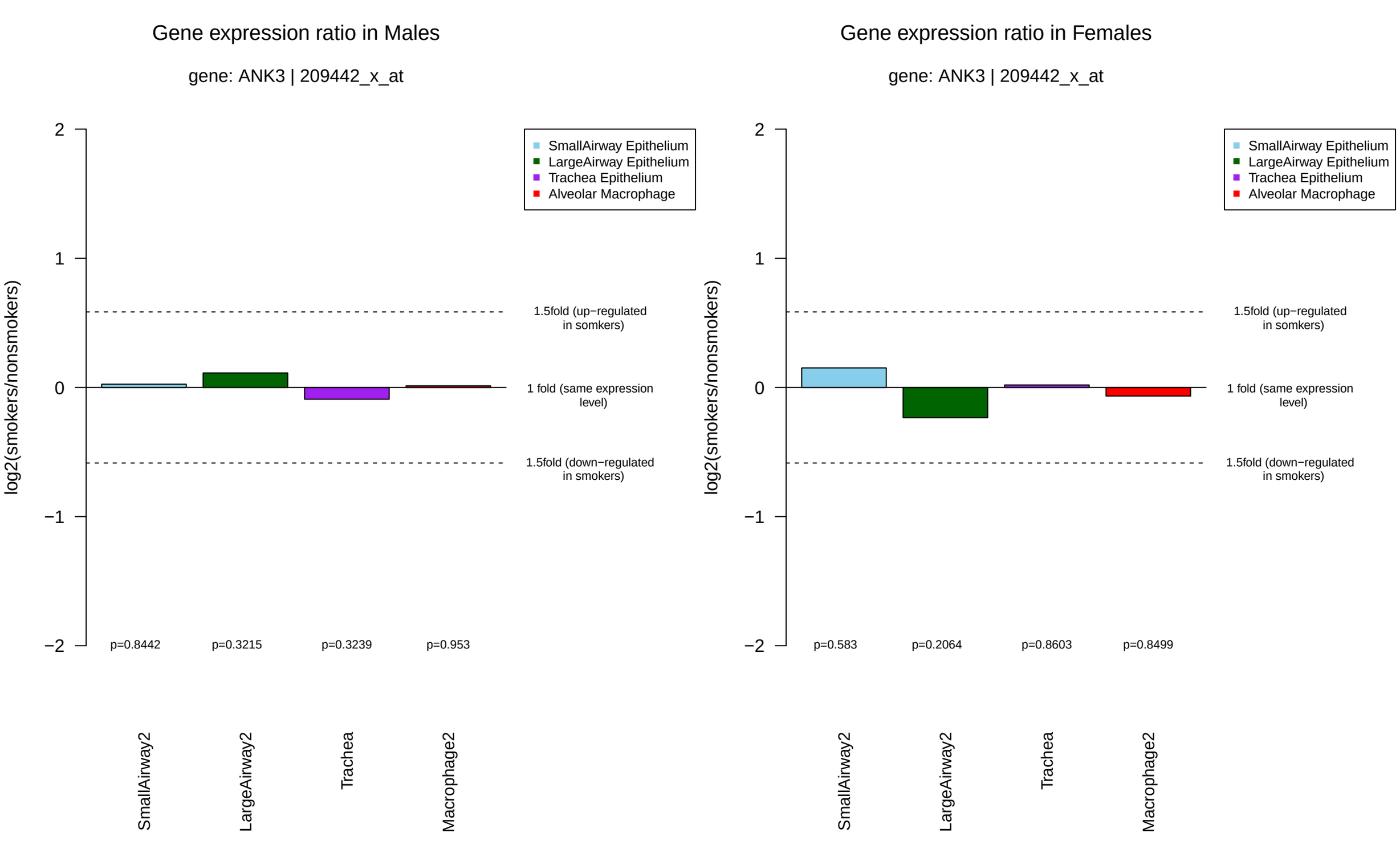
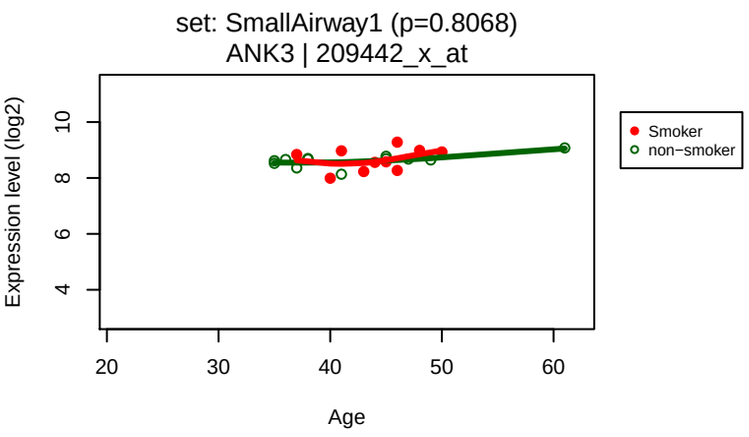
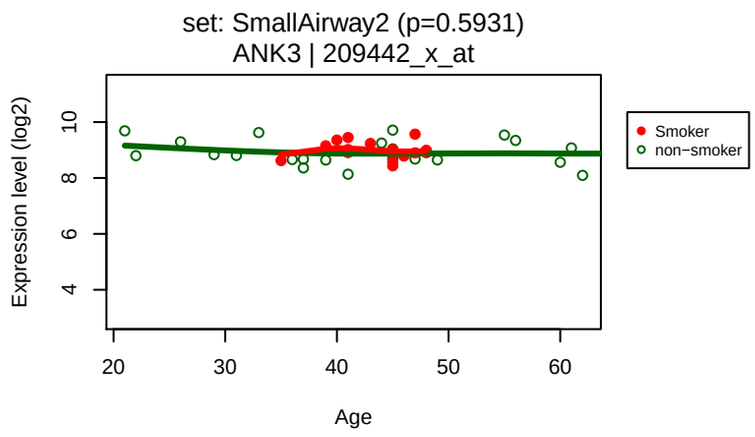
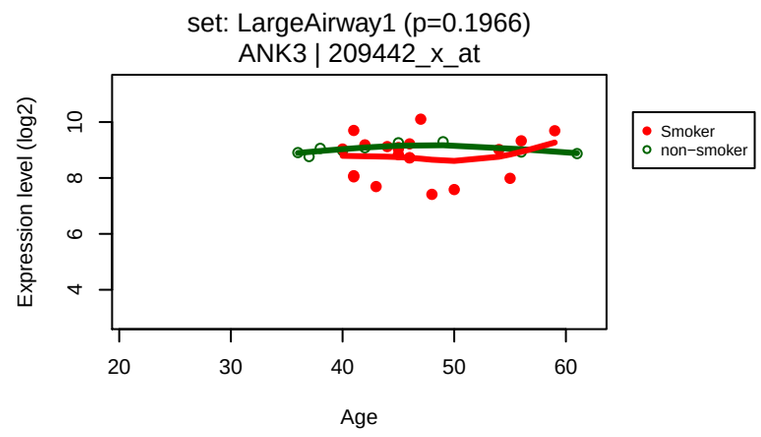

Gene expression of smokers and nonsmokers
gene: ANK3 | 209442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8068
p=0.5931
p=0.3892
p=0.1966
p=0.8716
p=0.2718
p=0.9897
p=0.8864
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

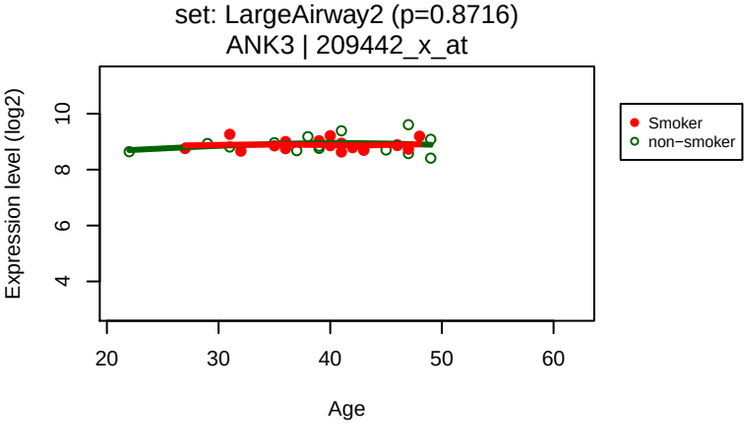

Gene expression ratio (smokers/nonsmokers)
gene: ANK3 | 209442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8068
p=0.5931
p=0.3892
p=0.1966
p=0.8716
p=0.2718
p=0.9897
p=0.8864
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

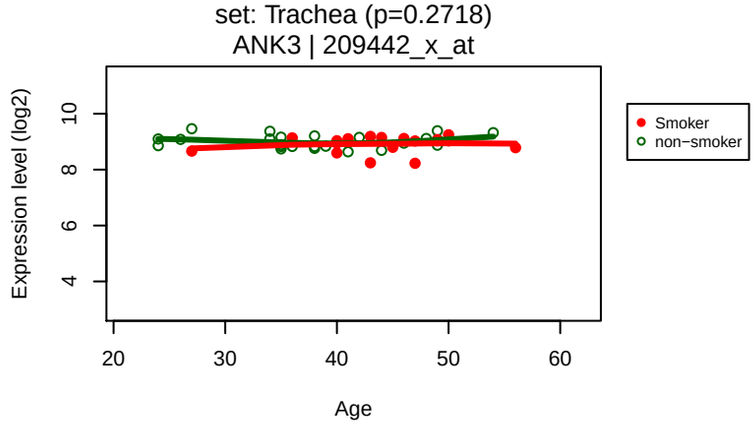

Gene expression ratio in Males
gene: ANK3 | 209442_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8442
p=0.3215
p=0.3239
p=0.953
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANK3 | 209442_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.583
p=0.2064
p=0.8603
p=0.8499
SmallAirway2
LargeAirway2
Trachea
Macrophage2

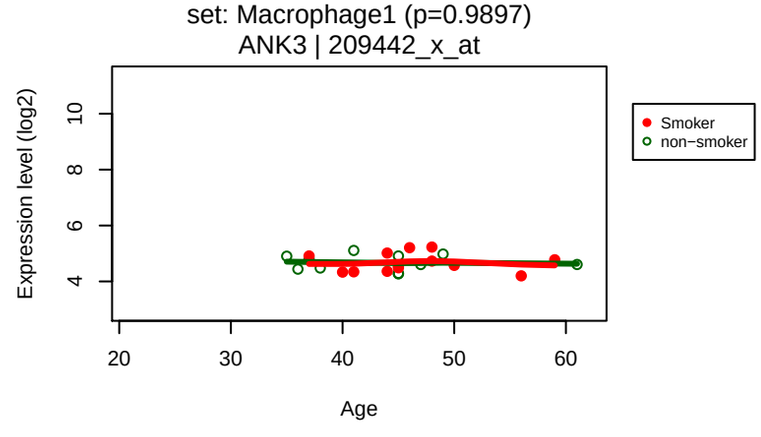

set: SmallAirway1 (p=0.8068)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker

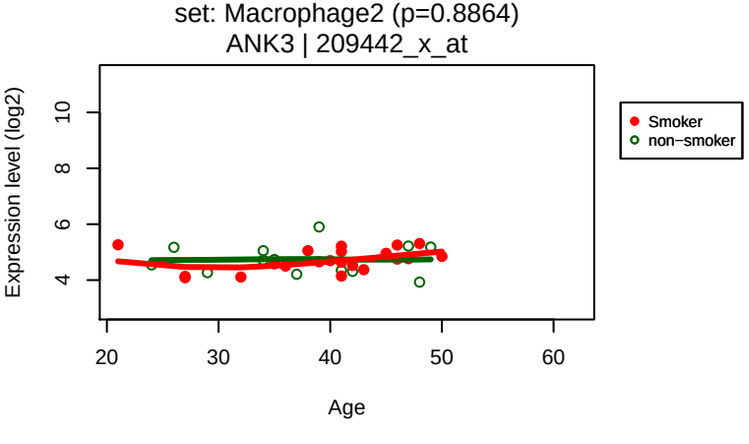

set: SmallAirway2 (p=0.5931)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker

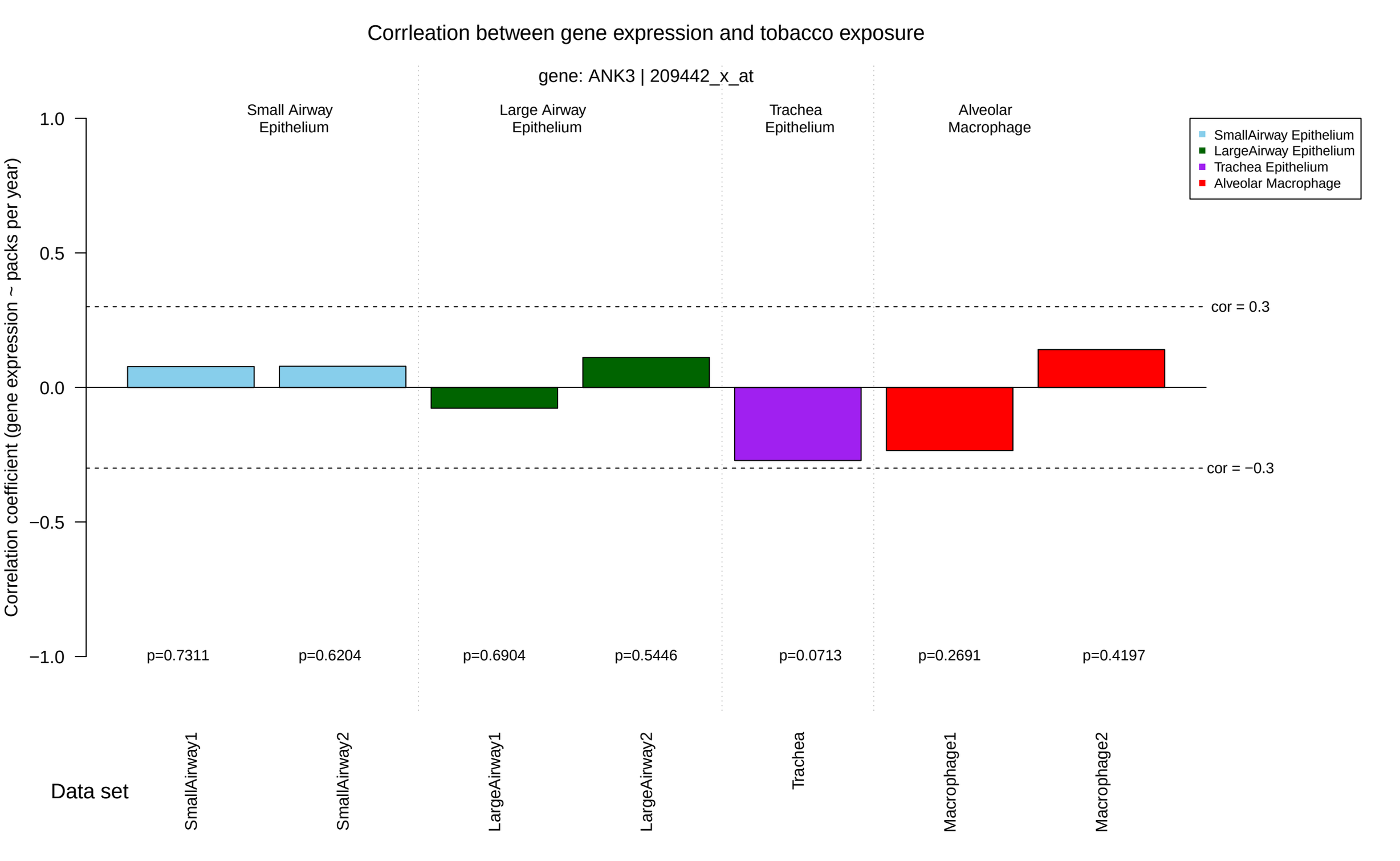

set: LargeAirway1 (p=0.1966)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8716)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2718)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9897)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8864)
ANK3 | 209442_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANK3 | 209442_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7311
p=0.6204
p=0.6904
p=0.5446
p=0.0713
p=0.2691
p=0.4197
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2